Bluestock Mutual Fund Analytics Capstone Project

Executive Summary

This project builds a comprehensive mutual fund analytics platform by integrating data engineering, financial analytics, and business intelligence methods. A unified analytical framework was created by integrating multiple datasets that include mutual fund schemes, historical NAVs, benchmark indices, investor transactions, portfolio holdings, and industry statistics.

The project encompasses ETL pipeline development, exploratory data analysis, assessment of fund performance, risk evaluation, analysis of investor behavior, and dashboard creation. To evaluate the fund's performance and risk profile, advanced financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were computed.

A Power BI dashboard with interactive features was created to offer insights into industry growth, fund performance, investor behavior, and market trends. The final solution shows how analytics can aid in making well-informed investment choices and enhance the understanding of financial data.

1. Project Objectives

The objectives of this project were:

- Create a structured ETL pipeline to process mutual fund datasets.
- Conduct exploratory data analysis to uncover industry trends.
- Assess fund performance based on financial metrics.
- Evaluate investment risk through statistical methods.
- Analyze investor behavior and SIP trends.
- Create a framework for recommending funds.
- Build an interactive dashboard using Power BI.
- Evaluate the risk of portfolio concentration by analyzing sector allocation data.
- Provide practical business insights via reporting and visualization.

2. Business Problem Statement

The mutual fund industry produces substantial amounts of performance, transaction, and portfolio data. Gaining meaningful insights from these datasets demands a systematic analytical method.

This project tackles the challenge by creating a unified analytics platform that can assess fund performance, detect investment risks, analyze investor behavior, and deliver insights via interactive dashboards and reports.

3. Scope of the Project

The scope of this project includes:

- Ingesting data from various mutual fund datasets.
- Cleaning and transforming data.
- Analysis of historical net asset value.
- Assessing performance across various schemes.
- Risk analytics and evaluation of potential losses.
- Analysis of investor transactions.
- Assessment of SIP behavior.
- Analysis of portfolio concentration.
- Comparison against benchmarks.
- Creating an interactive dashboard.
- Final reporting and presentation.

The project focuses on analytical evaluation and does not provide live investment advice or real-time trading functionality.

4. Data Sources

The analysis drew on several datasets that encompassed various facets of the mutual fund ecosystem.

4.1 Mutual Fund Scheme Information

Includes details of the scheme, fund categories, risk ratings, expense ratios, and past return data.

4.2 Net Asset Value (NAV) History

Offers daily NAV records for computing returns, volatility, and risk.

4.3 Benchmark Index Data

Includes NIFTY 50 and NIFTY 100 index values for performance comparison and Alpha-Beta calculations.

4.4 Industry AUM Dataset

Includes industry-level growth metrics like assets under management and the number of schemes.

4.5 Investor Transaction Dataset

Includes SIP, purchase, and redemption records for analyzing investor behavior.

4.6 Portfolio Holdings Dataset

Offers security-level portfolio allocation and sector details to support concentration analysis.

4.7 SIP and Category Flow Data

Monitors category-level inflows and trends in market participation.

5. ETL Architecture and Data Pipeline

A structured ETL process was put in place to ready datasets for analysis.

Extract

Raw CSV files with scheme data, NAV history, benchmark indices, investor transactions, and portfolio holdings were gathered and saved in the project repository.

Transform

Data preprocessing included:

- Handling missing values
- Removing duplicates

- Standardising formats
- Converting date fields
- Creating analytical features

Derived metrics included:

- Daily Returns
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha and Beta
- Maximum Drawdown
- VaR and CVaR

Load

Processed datasets were saved in designated directories and utilized for analytical notebooks and dashboard creation.

The ETL framework enhanced data quality, consistency, and reusability across the entire project lifecycle.

6. Exploratory Data Analysis (EDA)

Performance analytics was carried out to assess fund returns, risk-adjusted performance, and how well the funds held up during market volatility.

Daily Return Analysis

Daily returns were derived from historical NAV data and formed the basis for further risk and performance assessments.

CAGR Analysis

CAGR values over one-year three-year, and five-year periods were calculated to evaluate and compare long-term growth performance among the schemes.

Sharpe Ratio

The Sharpe Ratio quantifies excess returns in relation to the overall volatility of a portfolio, facilitating the ranking of funds based on risk-adjusted performance.

Sortino Ratio

Sortino Ratio This metric concentrated on downside risk and offered further insight into how efficiently an investment performed.

Alpha and Beta

Alpha and Beta were computed relative to benchmark returns to assess market sensitivity and the effectiveness of active management.

Maximum Drawdown

Maximum Drawdown quantified the most significant historical drop each fund experienced, serving as a measure of potential downside risk.

Fund Scorecard

A composite scoring system was created using metrics such as returns, Sharpe Ratio, Alpha, expense ratio, and drawdown to objectively rank funds.

Key Findings

- Several funds consistently ranked above their peers on multiple performance metrics.
- Strong returns did not always translate to solid risk-adjusted performance.
- Funds with positive alpha showed strong active management.
- Funds with lower drawdowns showed resilience during periods of market volatility.

7. Advanced Analytics and Risk Metrics

Advanced analytical techniques were put into place to evaluate portfolio risk, investor behavior, and diversification traits.

Value at Risk (VaR) and Conditional VaR

Historical VaR and CVaR were computed to assess possible downside losses under typical and stressed market conditions.

The generated VaR and CVaR report allows investors to compare risk profiles across different schemes and pinpoint investments that match their risk tolerance.

Key Observation: Funds that have delivered stronger historical returns did not necessarily carry lower downside risk, underscoring the need to assess both potential returns and risk levels when choosing investments.

Rolling Sharpe Ratio

A 90-day rolling Sharpe Ratio was calculated to track how risk-adjusted performance evolved over time.

The rolling Sharpe visualization offers useful insight into how consistently a fund performs and how stable its returns are, enabling investors to spot funds that show reliable long-term results instead of fleeting short-term wins.

Key Observation: Funds that maintained consistently positive Rolling Sharpe Ratios tended to show stronger risk-adjusted returns over time and were more capable of withstanding market volatility.

Investor Cohort Analysis

Investors were categorized by their year of first investment to analyze participation patterns and investment behavior across different groups.

The analysis also pinpointed the most commonly chosen schemes in each group, offering insights into evolving investor preferences and market trends.

Key Observation: Investors who joined the market earlier generally held higher cumulative investment values than more recent entrants, highlighting the advantages of long-term involvement.

SIP Continuity Analysis

Investor patterns of SIP contributions were reviewed to pinpoint those likely to stop their systematic investments.

Identifying at-risk investors can help fund houses implement targeted communication and retention strategies to encourage continued participation.

Key Observation: Consistent SIP participation remains a major driver of mutual fund growth, while early identification of irregular contributors can support investor retention initiatives.

Fund Recommendation Model

A straightforward recommendation framework was created using risk ratings and Sharpe Ratios to propose appropriate funds aligned with an investor's risk tolerance.

Sector Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was used to assess portfolio holdings, evaluating both diversification levels and sector concentration risk.

Portfolios with high concentration can deliver better returns when favored sectors thrive, but they also expose investors to greater risk during downturns specific to those sectors. Generally, diversified portfolios offer more stability and reduce the risk associated with overconcentration.

Key Observation: Funds having lower HHI scores showed greater diversification and were less vulnerable to market swings tied to specific sectors.

Key Findings

- Some funds exhibited notably greater downside risk compared to their peers.
- Performance adjusted for risk differed depending on market cycles.
- Earlier groups of investors provided a greater portion of the total invested capital.
- The majority of investors continued their regular SIP investments.
- The level of sector concentration varied significantly across equity-oriented funds.

8. Dashboard Overview

A Power BI dashboard was created to offer a unified overview of mutual fund industry trends, fund performance, investor behavior, and market analytics. The dashboard allows users to examine key metrics using interactive visualizations and filters.

8.1 Industry Overview Dashboard

Figure 8.1: Industry Overview Dashboard

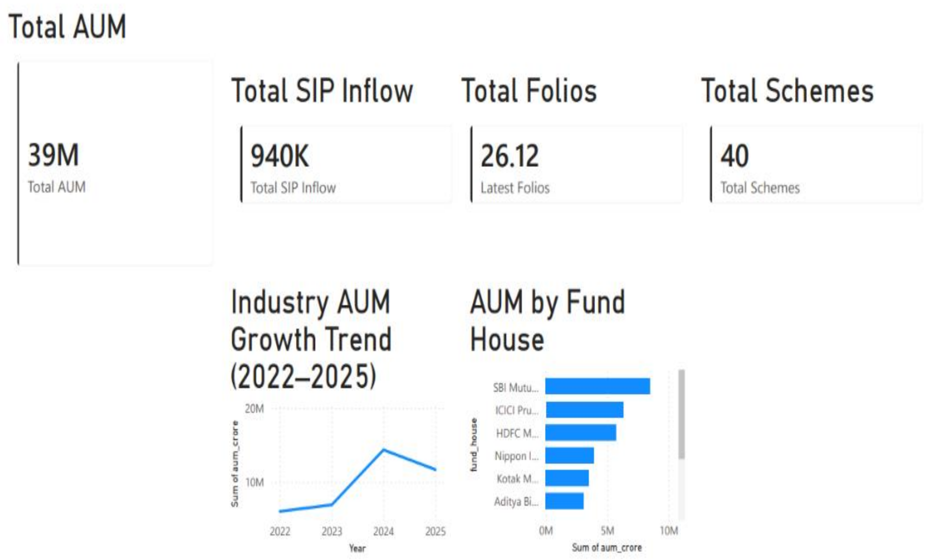
The Industry Overview dashboard displays key metrics that summarize the broader mutual fund ecosystem. Key Performance Indicators (KPIs) encompass Total Assets Under Management (AUM), SIP inflows, total folios, and total schemes.

The visualization of AUM growth underscores the expansion of the mutual fund industry from 2022 to 2025. There was a notable rise in AUM during the period, indicating rising investor involvement and stronger market confidence.

The fund house comparison chart allows stakeholders to evaluate asset accumulation among major asset management firms and pinpoint market leaders.

Insights

- Assets under management in the industry showed steady growth throughout the analysis period.
- SIP contributions kept rising, helping to fuel the industry's long-term growth.
- A few fund houses held a large portion of total assets.



8.2 Fund Performance Dashboard

Figure 8.2: Fund Performance Dashboard

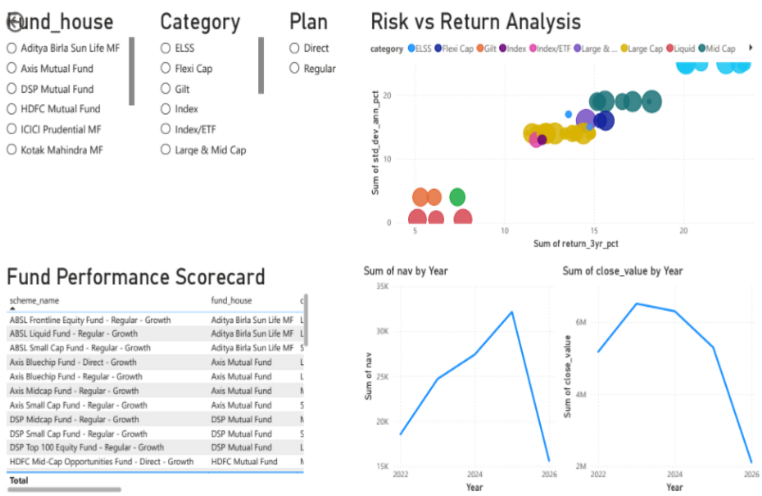
The Fund Performance dashboard allows users to compare mutual fund schemes based on return and risk metrics. Interactive filters enable users to analyze performance based on fund house, category, and investment plan.

The scatter plot of Risk vs Return shows how expected returns correlate with portfolio volatility across various fund categories. The size of the bubble indicates the fund's scale, offering extra insight into investor preferences.

The scorecard table offers a unified overview of scheme performance, and the NAV and benchmark trend charts aid in comparing historical data.

Insights

- Higher returns were typically linked to greater volatility.
- Several strategies achieved competitive risk-adjusted returns, even with moderate overall gains.
- Fund performance differed markedly depending on category and investment approach.



8.3 Investor Analytics Dashboard

Figure 8.3: Investor Analytics Dashboard

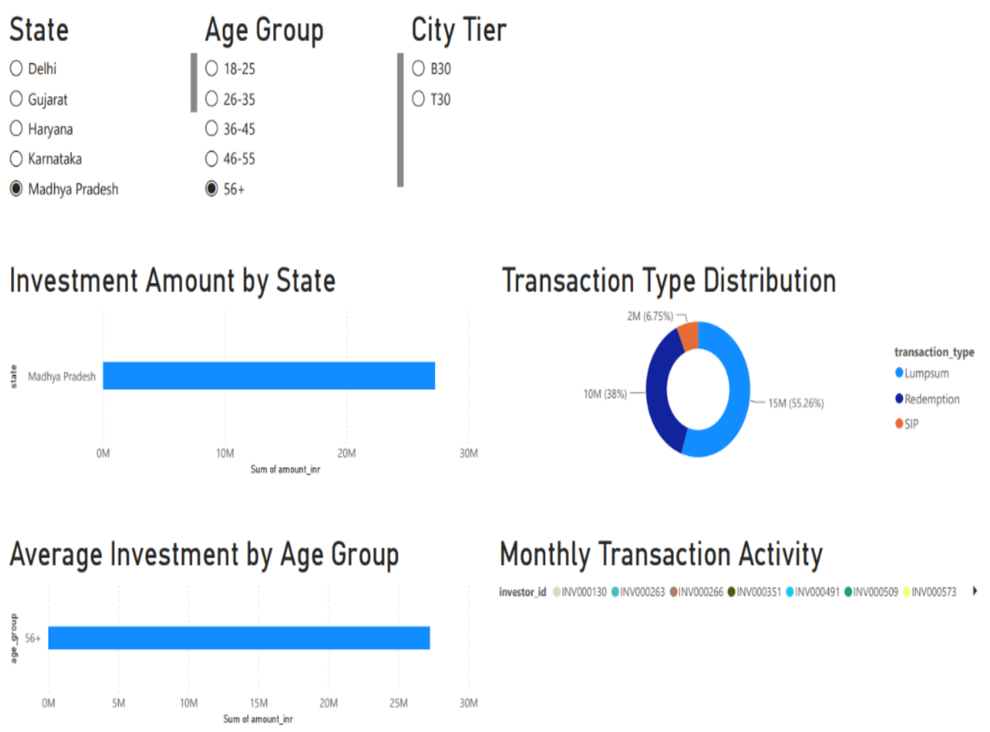
The Investor Analytics dashboard highlights transaction behavior and demographic engagement. Users have the option to filter results by state, age group, and city tier to analyze regional investment trends.

The visualizations cover investment amounts by state, the distribution of transaction types, average investments per age group, and monthly transaction activity.

The transaction distribution chart shows how SIPs, lump-sum investments, and redemptions each contribute relative to one another, offering a view into investor behavior.

Insights

- Investment involvement varied by region and demographic group.
- Ongoing investment activity played a major role in driving total transaction volume.
- In the analyzed sample, older age groups showed higher average investment values.



8.4 SIP and Market Trends Dashboard

Figure 8.4: SIP and Market Trends Dashboard

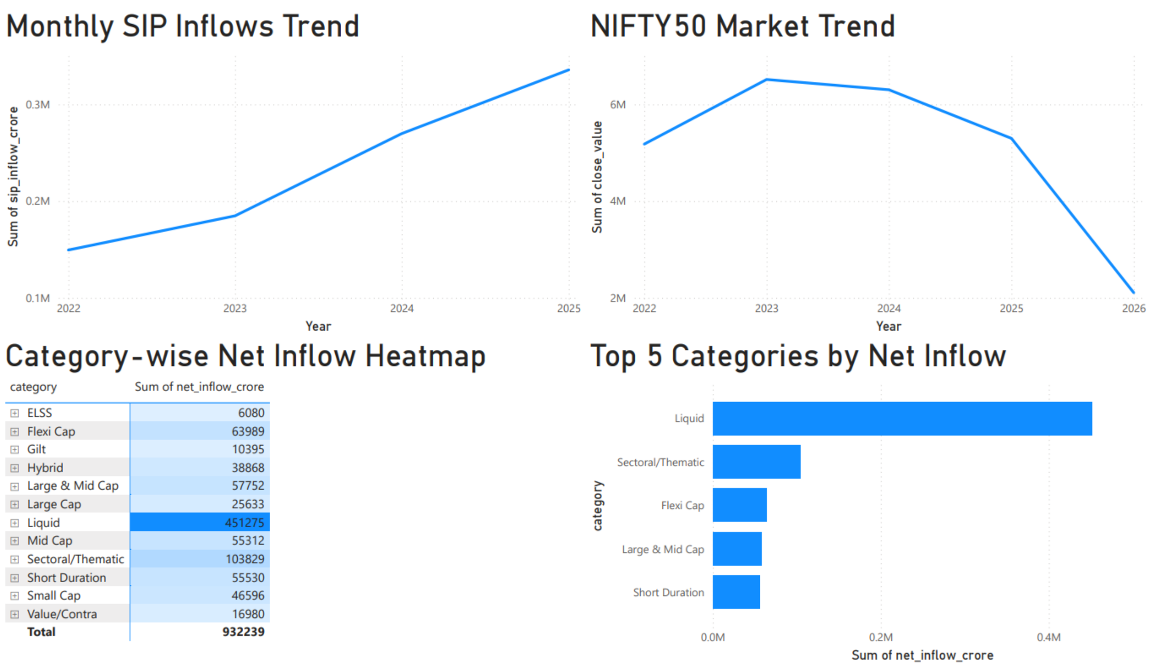
This dashboard integrates SIP inflow analysis with benchmark market trends to assess how investor behavior compares to market performance.

The SIP trend chart shows how systematic investments have grown over time, while the NIFTY 50 trend chart offers insight into broader market performance. An analysis of inflows by category reveals investor preferences among different mutual fund types.

The Top Categories chart highlights the categories that saw the greatest net inflows throughout the study period.

Insights

- SIP participation rose steadily throughout the analysis period.
- Changes in the market affected how investors distributed their investments among different categories.
- Some categories consistently drew greater net inflows compared to others.
- Investment behavior over the long term stayed strong even as markets shifted.



Dashboard Business Value

The dashboard converts intricate financial data into a user-friendly tool to support decision-making. By using interactive filtering and visual analytics, users can:

Monitor industry growth trends.

- Compare fund performance.
- Analyse investor behaviour.
- Evaluate market sentiment.
- Support data-driven investment decisions.

The dashboard serves as the primary visual interface of the project and demonstrates the practical application of business intelligence techniques within the mutual fund domain.

9. Key Findings

The project yielded multiple valuable insights in areas such as performance, risk, and investor analytics.

Key Finding 1

The mutual fund industry showed steady growth in Assets Under Management, indicating rising investor involvement and confidence.

Key Finding 2

Equity-focused funds saw the most investor attention and typically provided better long-term returns than other types.

Key Finding 3

Risk-adjusted performance differed significantly among schemes, showing that higher returns do not necessarily reflect improved investment efficiency.

Key Finding 4

Analysis of investor cohorts revealed that earlier groups accounted for a greater portion of total investments and sustained higher levels of participation.

Key Finding 5

Analysis of sector concentration showed notable variations in diversification across funds, with certain schemes displaying markedly greater concentration risk.

10. Project Limitations

While the project met its main goals, some limitations must be taken into account when evaluating the results.

Historical Data Dependency

The analysis relies completely on historical mutual fund and market data. Future fund performance could vary considerably as a result of shifting market conditions, economic developments, and investor attitudes.

Limited Dataset Coverage

The project makes use of a chosen group of mutual fund schemes and investor transaction records. The findings might not completely reflect the whole mutual fund industry.

Simplified Recommendation Model

The fund recommendation system employs a rule-based method that considers risk grades and Sharpe Ratios. It fails to consider individual financial goals, investment timeframes, tax implications, or shifting market conditions.

Absence of Real-Time Data

All calculations and dashboard visualizations are based on static datasets. The project did not include real-time NAV updates or live market feeds.

Assumptions in Risk Metrics

Risk metrics like VaR and CVaR rely on past return patterns and may fail to reliably forecast rare market shocks or upcoming financial crises.

Dashboard Scope

The dashboard was mainly created for analytical and educational use. Features like automated alerts, advanced forecasting, and user authentication were outside the scope of this implementation.

Although limited in scope, the project offers meaningful insights into mutual fund performance, investor behavior, and risk profiles, while showcasing how analytics can be applied in finance.

11. Recommendations

Based on the analysis performed, the following recommendations are proposed:

For Investors

- Evaluate funds using both return and risk-adjusted metrics.
- Maintain diversified portfolios across sectors and fund categories.
- Monitor fund concentration risk before making investment decisions.
- Continue disciplined SIP investments to support long-term wealth creation.

For Fund Managers

- Monitor downside risk metrics such as VaR and Maximum Drawdown.
- Improve diversification in highly concentrated portfolios.
- Use investor behaviour insights to strengthen retention strategies.

For Future Development

- Integrate real-time market and NAV data.
- Implement machine learning-based recommendation models.
- Develop predictive analytics for fund performance forecasting.
- Expand dashboard capabilities with additional investor segmentation features.

12. Future Scope

Although the current solution offers full mutual fund analytics capabilities, there are several improvements that can be introduced in future versions.

Real-Time Data Integration

Future updates may incorporate live NAV and market data via APIs, allowing for real-time tracking of fund performance and market trends.

Machine Learning-Based Recommendations

The existing recommendation system relies on rule-based logic. Machine learning models could be used to create tailored fund suggestions by analyzing investor profiles, past behavior, and market trends.

Predictive Analytics

Predictive analytics can involve building time-series forecasting models to anticipate future NAV changes, SIP inflows, and investment trends across categories.

Portfolio Optimization

Advanced methods like Modern Portfolio Theory (MPT) can be applied to suggest well-diversified investment approaches.

Cloud Deployment

Deploying the full analytics pipeline on cloud platforms enhances scalability, accessibility, and automated reporting features.

13. Conclusion

This project successfully created a complete mutual fund analytics platform by combining data engineering, financial analysis, and business intelligence methods.

Using a structured ETL pipeline, various datasets were converted into an analytical environment designed to facilitate performance evaluation, risk assessment, analysis of investor behavior, and measurement of portfolio concentration.

Using advanced metrics like Sharpe Ratio, Sortino Ratio, Alpha, Beta, VaR, CVaR, and HHI allowed for a more comprehensive insight into mutual fund traits than traditional return analysis alone.

The Power BI dashboard offered an interactive way to explore industry trends, fund performance, and investor activity, enhancing accessibility for business users and decision-makers.

Overall, the project illustrates how analytical techniques can be used on financial data to produce useful insights and aid in making well-informed investment choices.

14. References

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